

In T. 871/08 of 8 December 2011 prior art document D9 related to a method for hydrogen peroxide production by an anthraquinone method. The respondent had argued that the claimed subject-matter was novel, because the value 2.996 was lower than the lower limit of 3:1 defined in claim 1 at issue. This argument was not accepted by the board. When comparing a value from the state of the art (the value "2.996") with those claimed (the range of values "from 3:1 to 9:1"), the state of the art value had to be given the same accuracy as the one claimed. In the case at issue, the values in the claims had been quoted without any digit after the comma, which meant that for comparison purposes, the value 2.996 had to be rounded up to 3, which thus fell into the range of the claimed values (see, in particular, T. 1186/05, T. 708/05).

In T. 686/96 claim 1 related to a composition with a feature (iv) requiring a Perspex® abrasion value (PAV) in the range from about 12 to about 20 PAV. A prior art document disclosed in example 2 a composition having features (i) to (iii) of claim 1. With respect to novelty it had to be decided whether the known composition also had an abrasion value as required by feature (iv) of claim 1. The board established that the abrasion value of the known composition was somewhat below the lower value indicated in feature (iv) of the claim. Since the lower limit in the claim 1 was defined as "about 12", some interpretation was necessary. The board held that, when deciding on the novelty of the subject-matter of a claim, the broadest technically meaningful interpretation of a claim should be taken into account. In the board's view the scope of claim 1 was to be construed to mean that the indicated lower limit corresponded to the value disclosed in the prior art. Claim 1 was then considered to lack novelty. By contrast, in T. 2437/13 (nucleic acid sequences), the degree of sequence identity as defined in claim 1 represented the ratio (expressed in %) of identical nucleotides between a query nucleotide sequence that is pairwise aligned with a target nucleotide sequence, and the total number of aligned bases. Both numbers were integers which could be counted with absolute precision. Two bases either matched or did not. There was no error possible in calling two bases a match or not. The counting of matches could be repeated and would have led to the exact same result every time. There was no error of measurement or estimation and no error propagation in the calculated percentage sequence identity and hence no reason for rounding up the calculated ratio when it came to calculating the degree of sequence identity.

5.2.3 Difference in parameters

In T. 1764/06 the board noted that the only expressed distinction between the definition of the claimed photocatalyst and the photocatalyst of D1 relied on a parameter feature that was not usual, let alone common, for the skilled person. The onus of proof in that respect lay on the applicants, but no proven distinction had been shown. As regards the benefit of the doubt, which in the applicant's view was to be accorded to applicants in complex situations without direct disclosure, the position of the board was that in a situation where the applicants have used an unusual parameter feature to define their product, which unusual parameter feature represents the only distinction over otherwise identical known products, and the applicants have decided not to provide evidence that the parameter feature as such represents a difference in the claimed products from the known products, no benefit of the doubt could be accorded (see also T. 1764/06, T. 1995/15). The same applies in inter partes proceedings (T. 131/03, T. 740/01, T. 1666/16).

See also chapter III.G.5.2.2 "Cases in which the burden of proof was reversed".

5.2.4 Difference in composition

In T 80/96 (OJ 2000, 50) an aqueous solution of the claimed tartrate compound was described in the prior art. The board held that, in the case of an active agent which was known as such to be water-soluble, it was clear to a person skilled in the art that describing and claiming the active agent as a solution did not add to or change the definition of that active agent. Without further specification, the mere characterisation of a solvent or diluent as liquid or solid in a claim did not change the assessment of the novelty of the subject-matter of the claim. Analogously, in a claim directed to a preparation of a known structurally defined active agent with at least one auxiliary substance, in which the feature "with an auxiliary substance or auxiliary substances" meant that something was added to the active agent, the admixture of an unspecified auxiliary substance could not, in view of the unlimited number of substances which might enter into consideration, be deemed a substantive and distinctive addition to the active agent, unless this feature, which was necessary if novelty was to be acknowledged, was specified in such a way that a person skilled in the art could recognise what it was that should be added to the active agent. The claim was therefore not new.

5.2.5 Functional features

In T 500/89 it could only be seen from the disclosure considered in its entirety that the prior art document did not deprive the claimed invention of novelty, because the method constituting the closest prior art differed from the claimed method in one functional characteristic. The disputed patent related to a method for the production of photographic material by the simultaneous application of several layers of fluid photographic coating materials. Although the document cited in support of the opposition listed the numerical ranges for layer thickness, viscosity, coating speed, etc. used in the method claimed, the latter was nevertheless held to be new because the cited document described the choice of these numerical ranges as leading to intermixing between two particular layers. The contested patent was to be assessed according to a different criterion because it described the application of the layers as being "substantially free from intermixing". The "intermixing" described as an objective in the citation was not merely a stated purpose not constituting one of the technical features of the method described, but a functional feature – a criterion, in effect – forming an essential element of the teaching set out in this publication.

According to the established case law, where a claim concerns an apparatus differing from a known apparatus only as regards to the use indicated, then the use is not an apparatus feature, meaning that two apparatuses (differing only in the intended use) are identical in terms of structure. If the known apparatus is suitable for the claimed use, the claimed invention lacks novelty. If, however, the claim is for a process, the situation is not comparable. In such a case, the use feature is a functional process feature comparable in category with the other features (steps) of the process (T 848/93 cited by T 1931/14). The opponent in T 1931/14 also referred to decision T 304/08, but the board stated that the situation in T 304/08 was quite different. The board in T 1931/14 concluded that T 848/93

and T.304/08 demonstrated that in the context of a method it was important to differentiate between different types of stated purpose, namely those that define the application or use of a method, and those that define an effect arising from the steps of the method and implicit therein. Where the stated purpose defines the specific application of the method, in fact it requires certain additional steps which are not implied by or inherent in the other remaining steps defined in the claim, and without which the claimed process would not achieve the stated purpose. In this manner the stated application represents a genuine technical limitation of the method and the claimed method must be applied in that manner. On the other hand, where the purpose merely states a technical effect which inevitably arises when carrying out the other remaining steps of the claimed method and is thus inherent in those steps, such a technical effect has no limiting effect because it is not suitable for distinguishing the claimed method from a known one. In T.1930/14 the board considered that this finding in T.1931/14 (point 2.2.4 of the Reasons) could only hold in cases where it was unambiguously clear that the purpose implied such steps and where it was also unambiguously clear what those steps in fact were. In T.1930/14, the board could identify no indication, either in the claim itself or in the description, that would lead the skilled person to understand that the stated purpose "for purifying a monoclonal antibody from aggregates thereof" implied that the claimed method contained additional steps. See also T.1399/16.

In T.872/09 the board stated that the claimed sensor was defined by reference to characteristics of its response when used in a measurement set-up. Since none of the determining aspects of the measurement set-up was defined in claim 1, the technical features of the claimed sensor which were responsible for providing the measurement referred to in the claim remained obscure. Legal certainty required that a claimed subject-matter cannot be regarded as novel over the prior art on the basis of an ambiguous feature (see T.1049/99). Hence, defining a functional feature of the claimed electrochemical sensor under undefined operating conditions was not appropriate to provide any distinction of the claimed subject-matter over the prior-art electrochemical sensors.

T.2170/13 dealt with the re-drafting of claim 1 as a method claim (auxiliary request). The board endorsed the approach in T.1822/12 that G.2/88 (in relation to functional technical features) related only to a use claim, namely to a claim for the new use of a known compound, that the case law had constantly interpreted G.2/88 in a very restrictive manner and that there was no possibility to expand the ruling in G.2/88 and G.6/88 to a claim worded otherwise. Hence the undisclosed purpose could not be considered to be a distinguishing functional technical feature of the claim. Following long technical arguments, the board also concluded (on the main request) that the final claimed feature was implicitly disclosed in prior art D1.

The term "means" is frequently used in the plural in the wording of patent applications, especially for functional definitions, but this does not actually rule out that there is only a single structural element (T.1794/12).

5.2.6 Generic disclosure

It is established case law that a specific disclosure destroys the novelty of a generic feature in a claim but that a generic disclosure does not destroy the novelty of a specific feature (see T 651/91, T 776/07, T 1174/05, T 6/04, T 776/07). A generic disclosure does not normally deprive a claimed specific embodiment of novelty (T 1786/09).

In T 651/91 the board pointed out that a generic disclosure did not normally deprive any specific example falling within that disclosure of novelty. A disclosure could be generic even where it only left open the choice between two alternatives. In T 508/91 the board held that, on the other hand, the prior disclosure of the subset "vegetables" deprived the wider set "fruits and plants" of novelty (see also T 1204/00).

The disclosure "rastered halftone images, such as portraits" of prior art document D3 necessarily included portraits which fall under the understanding of the feature "at least one digitised mark made of at least one set of dots appearing as a three dimensional mark when viewed in transmitted light" as defined in the patent in suit. The fact that, in addition, there might be portraits which fall under the disclosure of document D3 but which did not correspond to the claimed feature did not invalidate the existence of those that do: the wording of the claimed feature was even more generic than the disclosure in document D3 and was thus anticipated by it (T 687/14).

5.2.7 Product claim with process features

In T 815/93 and T 141/93 the claims comprised both product features and features for a process for manufacturing the product. In both cases, only the process features distinguished the invention from the prior art. Following the case law on the novelty of product-by-process claims, the board found that process features not previously described could establish the novelty of the claimed product only if they caused it to have different properties from the products previously described. Neither the patent proprietor in the first case nor the applicant in the second case could demonstrate this.

In T 32/17 the board found that a process feature in a product-by-process claim only contributes to the novelty of a product claim insofar as it gives rise to a distinct and identifiable characteristic of the product (see also T 179/03). From hybridomas' deposit information the skilled person derived that hybridomas producing the claimed antibodies had been deposited and had been assigned deposit numbers. This deposit information however did not convey any technical information about the chemical composition or molecular structure of the antibodies produced by these hybridomas, such as their amino acid sequence, either explicitly or implicitly. The board thus concluded that the deposit of a hybridoma under R. 31 EPC for compliance with the disclosure requirement of Art. 83 EPC did not in itself convey any technical information about the molecular structure of the monoclonal antibody produced by said hybridoma, such as its amino acid sequence.

5.2.8 Non-technical distinguishing features

Whereas novelty is not necessary to establish the technical character of an invention, the converse is not true as novelty and inventive step can only be established on the basis of the technical features of the invention. This is in line with the case law of the boards of appeal (T 154/04, OJ 2008, 46).

The Enlarged Board of Appeal held in decision G 2/88 (OJ 1990, 93) that the claims of a European patent should clearly define the technical features of the subject invention and thus its technical subject-matter, in order that the protection conferred by the patent can be determined and a comparison can be made with the state of the art to ensure that the claimed invention is, inter alia, novel. A claimed invention lacks novelty unless it includes at least one essential technical feature which distinguishes it from the state of the art.

In T 154/04 (OJ 2008, 46) the board stated that it is legitimate to have a mix of technical and "non-technical" features appearing in a claim, in which the non-technical features may even form a dominating part of the claimed subject matter. Novelty and inventive step, however, can be based only on technical features, which thus have to be clearly defined in the claim. Non-technical features, to the extent that they do not interact with the technical subject matter of the claim for solving a technical problem, i.e. non-technical features "as such", do not provide a technical contribution to the prior art and are thus ignored in assessing novelty and inventive step.

In T 2050/07 the board confirmed the established case law according to which features that do not contribute to the technical character of an invention and do not interact with the technical subject-matter of the claim for solving a technical problem have to be ignored when assessing inventive step, such features should equally be ignored when assessing novelty. The board therefore examined whether or not the distinguishing features in the case before it made a technical contribution.

In T 2191/13 the board observed that novelty could not be established on the basis of a distinction **made only mentally**, i.e. derived solely from pre-existing knowledge and not reflected in the technical features of the claimed subject-matter. It cited the Enlarged Board's ruling in G 2/88 that a claimed invention lacked novelty unless it included at least one essential technical feature distinguishing it from the prior art. It followed from that decision that a feature of a merely mental nature could not be a new technical feature for the purposes of Art. 54 EPC and, therefore, that a claim containing only already known technical features could not be considered novel. This approach was also taken in T 959/98, T 553/02 and T 154/04.

6. Chemical inventions and selection inventions

6.1. General

The state of the art often includes documents containing technical teachings described in general terms; these teachings in turn subsume a number of more specialised technical teachings. In assessing the novelty of subject-matter that can be subsumed under a